PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Hyperparathyroidism
PROCEDURE: Not answered
SPECIFIC CLINICAL QUESTION: Not answered
OUTSIDE TISSUE DIAGNOSIS: Not answered
PRIOR MALIGNANCY: Not answered
CHEMO RADIATION: Not answered
ORGAN TRANSPLANT: Not answered
IMMUNOSUPPRESSION: Not answered
OTHER DISEASES: Not answered

UUID:A2821B1D-5D8B-45B0-8976-99A15E49B84E
TCGA-BJ-A0ZG-01A-PR
Redacted

ICD-O-3
carcinoma, papillary, follicular variant 8340/3
Site: thyroid C73.9
6/10/12

ADDENDA:

Addendum

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block 2A:

- **A. BRAF mutation POSITIVE (p.V600E, c.1799T>A).**
- **B. Mutations in *NRAS61, HRAS61, KRAS12/13* NOT identified.**
- **C. FISH test results are negative for *RET/PTC* rearrangement.**

NOTE:
DNA was extracted in the amount sufficient for testing.

BACKGROUND:
Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between *BRAF* V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, *BRAF* V600E mutation and *RET/PTC* and *PAX8/PPARg* rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). *RAS* mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. *Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. Am J Surg Pathol 2006, 30:216-222*
2. *Xing M, et al. BRAF mutation predicts a poorer clinical prognosis for papillary thyroid cancer. J Clin Endocrinol Metab 2005, 90:6373-6379.*
3. *Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. J Clin Endocrinol Metab 2008, 93:3943-3949.*
4. *Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential BRAF(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. Ann Surg Oncol. 2009 Feb;16(2):240-5.*
5. *Nikiforova MN, et al. RAS point mutations and5PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. J Clin Endocrinol Metab 2003, 88: 2318-26.*
6. *Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. Endocrine Pathology: Differential Diagnosis and Molecular Advances. Humana Press, Totowa, 2004, 191-209.*

MUTATIONAL ANALYSIS:
For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify BRAF codons 599-601, NRAS codon 61, HRAS codon 61, and KRAS codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by r sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA.

Addendum

Criteria	Yes	No
Diagnosis Discrepancy		X
Primary Tumor Site Discrepancy		X
HIPAA Discrepancy		X
Prior Malignancy History		X
Dual/Synchronous Primary Noted		X
Case is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials	Date Reviewed:	6/10/12

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block 1A:

- ***D.*** **Loss of heterozygosity IDENTIFIED at 1p, 10q, and 13q.**
- ***E.*** **Allelic loss 38% (5/13 informative loci).**
- ***F.*** **Loss of heterozygosity of 1p IDENTIFIED; Loss of Heterozygosity of 13q *(RB1)* IDENTIFIED.**

NOTE:
The quantity and quality of isolated DNA was acceptable for testing.

BACKGROUND:

Loss of tumor suppressor gene alleles is known to occur in parathyroid neoplasms and to correlate with tumor progression. As reported in the literature, parathyroid carcinomas frequently have a mean fractional allele loss of ~ 63%, and many of these tumors have retinoblastoma gene deletions at 13q14 (1,2). Parathyroid adenomas typically have lower frequency of LOH, frequently fewer than 11% (1). Some studies have shown that parathyroid adenomas and carcinomas often have identifiable LOH on chromosome 1p, while hyperplasia is less commonly affected by this genetic loss (1,3). However, not all tumors reveal such a correlation and, therefore, the LOH profile should be interpreted in the context of the cytologic and histologic findings and the patient's clinical history.

1. *Hunt J, et al. Am J Surg Pathol. 2005;29:1049-55.*
2. *Cetani F, et. al. Clinical Endocrinology. 2004;60:99-106.*
3. *Valimaki S, et. al. International Journal of Oncology. 2002;21:727-35.*

LOSS OF HETEROZYGOSITY ANALYSIS:

For cytology samples, extraction of DNA was performed from the fluid or sample provided. For surgical specimens, manual microdissection was performed to include neoplastic tissue and normal adjacent tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. DNA was isolated using standard laboratory procedure. Optical density readings were obtained. Sixteen microsatellite markers (D1S407 (p21,1p36.21), D1S1161 (p21, 1p35.1), D1S1183 (HRPT2, 1q25.3), D3S1516 (VHL, 3p25.3), D5S659 (APC, 5q23.2), D9S251)p16, 9p21.3), D10S520 (PTEN, 10q23.31), D10S1173 (PTEN, 10q23.31), D11S1319 (KAI1, 11p11.2) D11S4177 (11p15.5), D11S916 (MEN1, 11q23), D11S387 (MEN1, 11q25), D13S161 (RB, 13q14.3), D13S319 (RB, 13q14.3), D17S516 (p53, 17p13.1), D17S1844 (p53, 17p13.1), D17S1877 (17q21), D22S1150 (NF2, 22q12.2) that have been previously found to be involved in parathyroid neoplasia and co-localize with known tumor suppressor genes were used for analysis. PCR was performed using fluorescently labeled primers and the products of amplification were detected using capillary electrophoresis on) platform. Relative fluorescence was determined for individual alleles and the ratio of peaks was calculated ( ). Normal tissue was examined to determine whether the patient is heterozygous at the marker (genetically informative) and neoplastic tissue was then analyzed to detect loss of heterozygosity. Thresholds for significant allelic imbalance were determined using normal (non-neoplastic) specimens for every marker. Loss of heterozygosity was determined using (N^L / N^S)/(T^L/T^S) formula and reported when allelic ratio for a particular marker was below 0.5 or above 2.0. When normal tissue was not available, peak height ratios falling outside of 2 SDs beyond the mean for each polymorphic allele paring were assessed as showing loss of heterozygosity.

FINAL DIAGNOSIS:

PART 1: PARATHYROID, RIGHT SUPERIOR, EXCISION (948 MG) –
ENLARGED AND HYPERCELLULAR PARATHYROID GLAND WITH FIBROSIS, SEE COMMENT.

PART 2: THYROID, TOTAL THYROIDECTOMY –

- **A. PAPILLARY MICROCARCINOMA, 0.7 CM, LOCATED IN RIGHT LOBE, SEE COMMENT.**
- **B. NO ANGIOLYMPHATIC INVASION OR EXTRATHYROIDAL EXTENSION.**
- **C. ALL MARGINS ARE FREE OF TUMOR.**
- **D. BACKGROUND THYROID SHOWS NODULAR HYPERPLASIA WITH A DOMINANT HYPERPLASTIC NODULE, 2.0 CM, LOCATED IN LEFT LOBE.**

[redacted]

COMMENT:

Part 1: The intraoperative PTH level dropped from 512 pg/ml to 78 pg/ml after excision of the right superior parathyroid gland, which, in addition to the microscopic appearance, is consistent with an adenoma. The parathyroid gland additionally shows significant fibrosis, mostly in the peripheral location, but no definitive invasion into adjacent soft tissue. During surgery, this parathyroid gland was found to be mildly adherent to the right thyroid lobe. To provide additional information on the nature of this apparently adenomatous lesion, the molecular LOH test will be performed and reported as an addendum.

Part 2: The focus of papillary microcarcinoma shows follicular architecture, is encapsulated, and reveals no infiltrative growth. Molecular mutational test will be performed and reported as an addendum.

<u>Results</u>

RET/PTC Methodology Description:

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled SpectrumGreen) is a 207 kb DNA probe located at band 10q11.2. The H4 and 369 probes are approximately 100kb each (labeled in SpectrumOrange), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three RET signals are identified in greater than 8% of the analyzed cells.

RET /PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *PTC* (369/H4) SpectrumOrange and the *RET* SpectrumGreen probes.